

习题课

03.23

TA: 郑东泽 (陈树明组硕士)

工学院南楼338

12432617@mail.sustech.edu.cn

习题课

- 作业题目讲解
- 补充习题（选做）讲解
- 提问答疑

习题课可自愿前往以及随时离开

作业评分

请同学们认真对待作业

按时提交作业

整洁工整

Home Work 1

1. 原胞与单位晶胞

一个单晶在图1中以二维形式表示

(a) 列出该晶体的单位晶胞。

(b) 列出该晶体的原胞。

1. Primitive and unit cell

A single-crystal is represented two-dimensionally in Figure 1.

(a) List the unit cell(s) of the crystal.

(b) List the primitive cell(s) of the crystal.

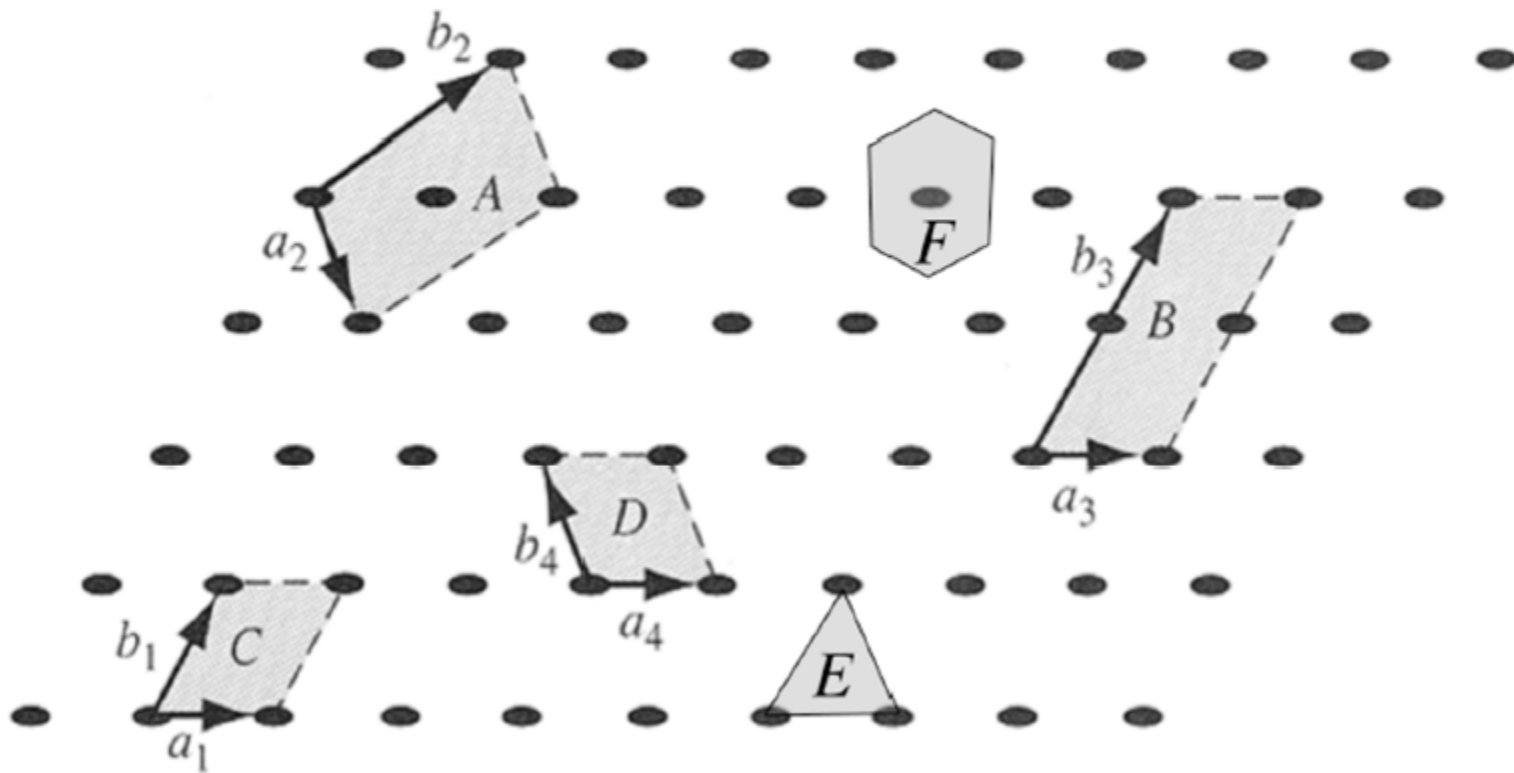


Figure 1: Two-dimensional representation of a single-crystal lattice

图1：单晶晶格的二维表示

Home Work 1

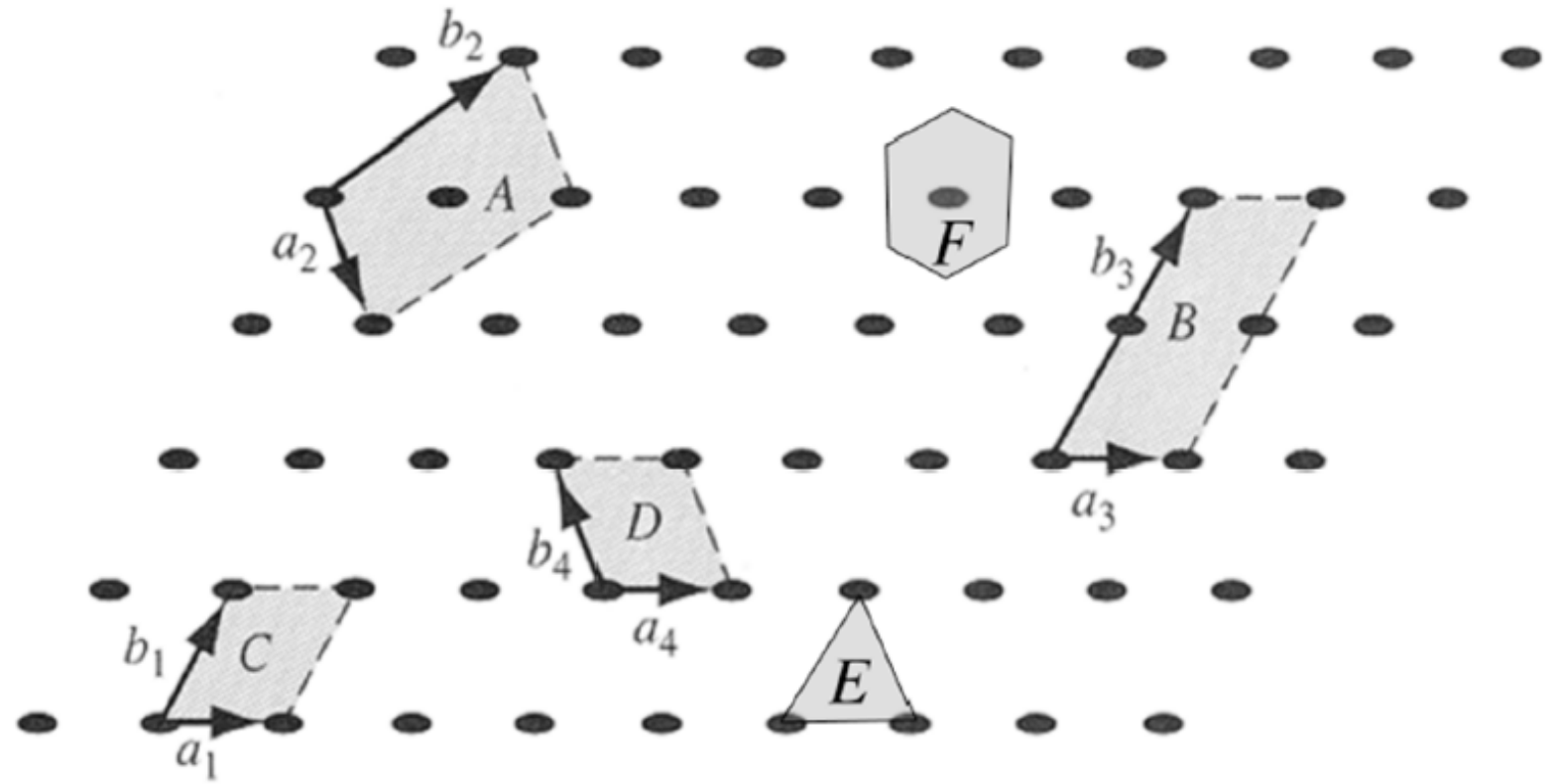
Answer:

(a) 列出该晶体的单位晶胞

A、B、C、D、F

(b) 列出该晶体的原胞

C、D、F



Unit cell : A small volume of a crystal that can be used to reproduce the entire crystal

A generalized primitive unit cell

A **PRIMITIVE CELL** is the *Smallest* unit cell!

Home Work 1

2. 确定单位晶胞中原子数

确定以下结构的单位晶胞中原子数: **Answer:**

(a) 面心立方晶格

(b) 体心立方晶格

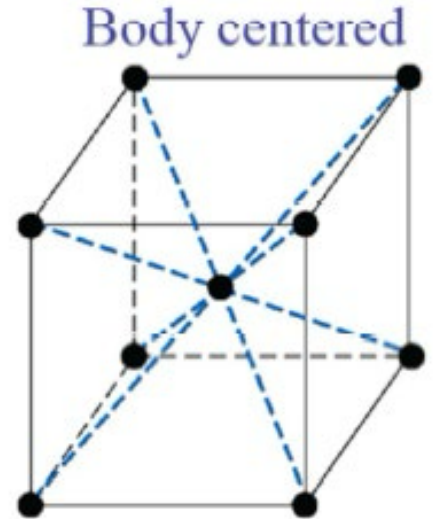
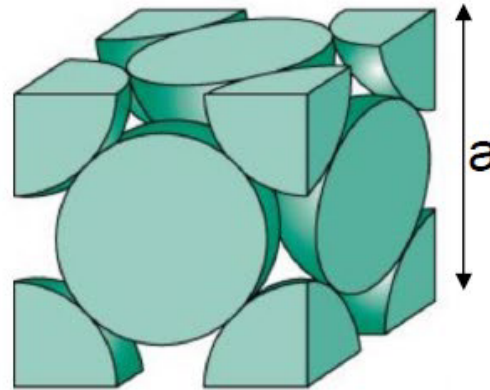
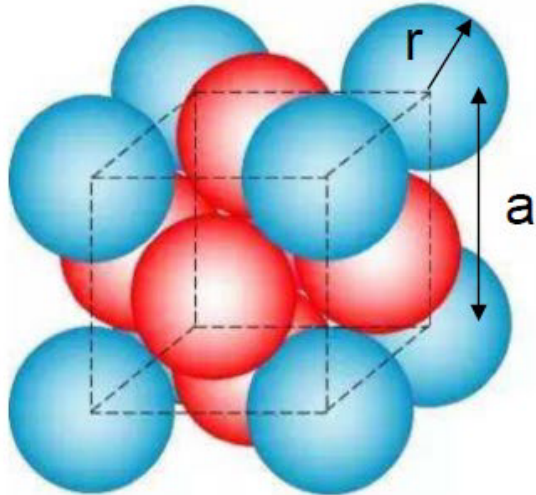
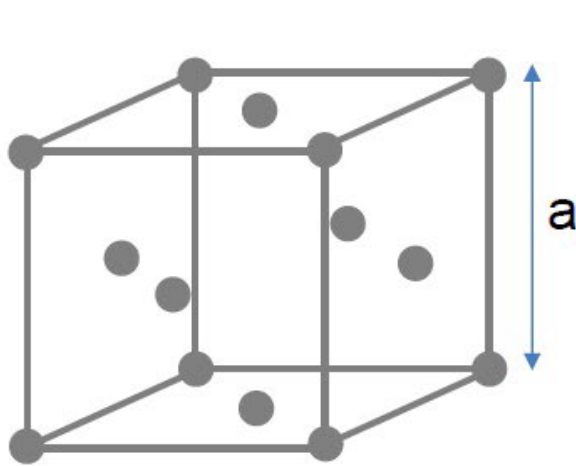
(c) 金刚石晶格

(a) face-centered cubic (fcc) unit cell

A sphere at corner of fcc unit cell shares the volume with 8 unit cells. Therefore the number of spheres at the corner is $\frac{1}{8}$. There is 8 corners in the fcc unit cell. A sphere at the face shares with 2 unit cells, i.e., the number is $\frac{1}{2}$. There are 6 faces in the fcc unit cell. The total number of the spheres is then $\frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$.

(b) body-centered cubic (bcc)

$$\frac{1}{8} \times 8 + 1 = 2$$



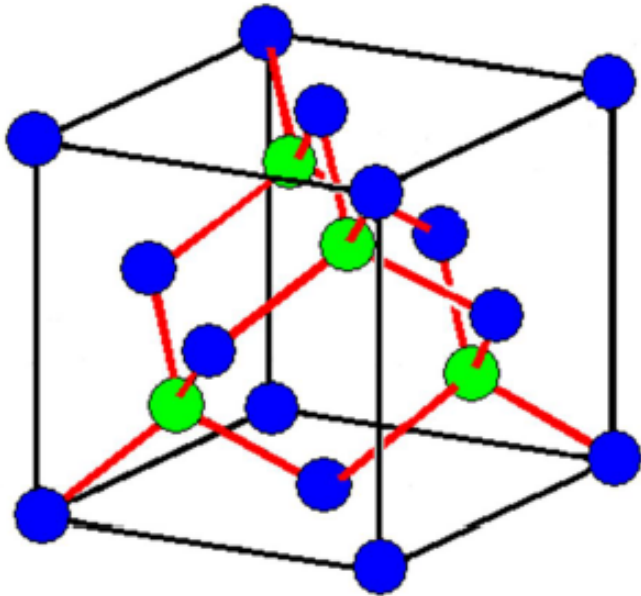
Home Work 1

Answer:

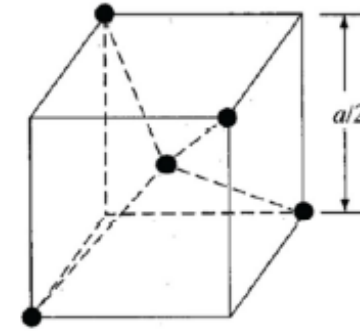
(c) diamond

$$\frac{1}{8} \times 8 + \frac{1}{2} \times 6 + 4 = 8$$

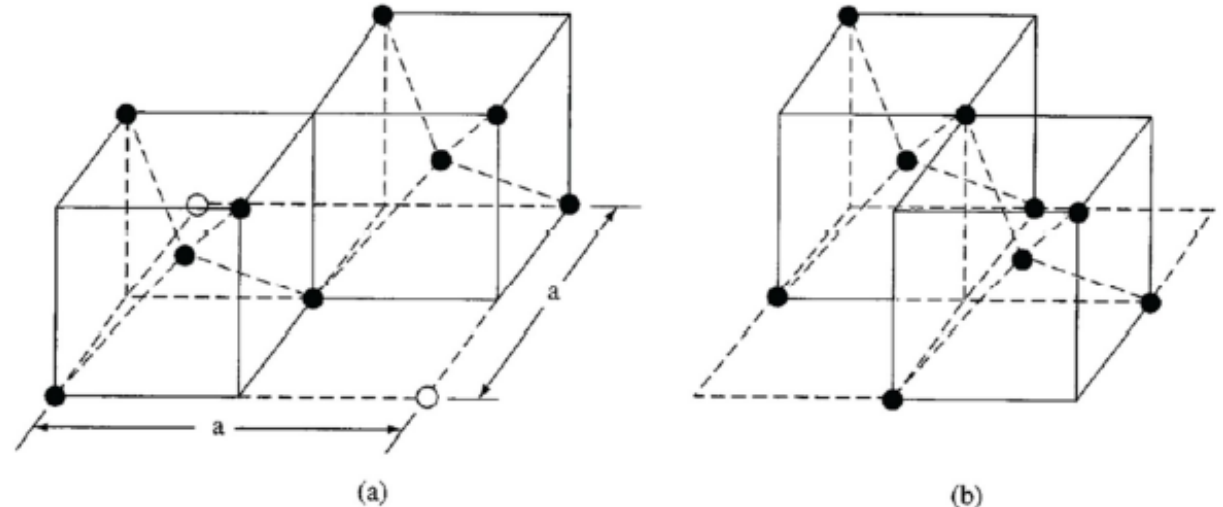
Note: The diamond unit cell consists of 4 bcc unit cells. Therefore the number of atoms is 4 times the bcc value.



- Blue: atoms of a fcc cell
- Green: new atoms introduced to a fcc cell
- Diamond unit cell = fcc cell + 4 new atoms



The tetrahedral structure of closest neighbors in the diamond lattice



Bottom half (a) and top half (b) portions of diamond lattice.

Home Work 1

3. 确定原子体密度

确定以下晶体结构的原子
体密度，晶格常数为 $a=5\text{\AA}$

(a) 简单立方晶格

(b) 体心立方晶格

(c) 面心立方晶格

$$a=5\text{\AA}=5\times 10^{-8}\text{ cm}$$

3. Determine the volume density of the atom in crystals with (a) simple-cubic, (b) body-centered cubic and (c) face-centered cubic crystal structures with a lattice constant $a=5\text{\AA}$.

Answer:

Density = $\frac{N}{V}$, where the N is the number of atoms in the unit cell and V is the volume of the unit cell.

$$(a) \frac{1}{a^3} = \frac{1}{(5\times 10^{-8})^3} = 8 \times 10^{21} [\text{atoms}/\text{cm}^3]$$

$$(b) \frac{2}{a^3} = \frac{1}{(5\times 10^{-8})^3} = 1.6 \times 10^{22} [\text{atoms}/\text{cm}^3]$$

$$(c) \frac{4}{a^3} = \frac{4}{(5\times 10^{-8})^3} = 3.2 \times 10^{22} [\text{atoms}/\text{cm}^3]$$

原子体密度的计算如下：

$$\text{体密度} = \frac{\text{每晶胞的原子数}}{\text{每晶胞的体积}}$$

Home Work 1

4. 计算原子密度

(a) GaAs 的晶格常数为 $5.65\text{\AA}$

确定每立方厘米中 Ga 原子和 As 原子的数量。

(b) 确定锗半导体中锗原子的体密度。

锗的晶格常数为 $5.65\text{\AA}$ 。

4. (a) The lattice constant of GaAs is $5.65\text{\AA}$. Determine the number of Ga atoms and As atoms per cm^3 . (b) Determine the volume density of germanium atoms in a germanium semiconductor. The lattice constant of germanium is $5.65\text{\AA}$.

Answer:

(a) GaAs has the zincblende structure and has 4 Ga atoms per unit cell.
$$\text{Density} = \frac{4}{(5.65 \times 10^{-8})^3} = 2.22 \times 10^{22} [\text{atoms}/\text{cm}^3]$$

The same for the As atoms.

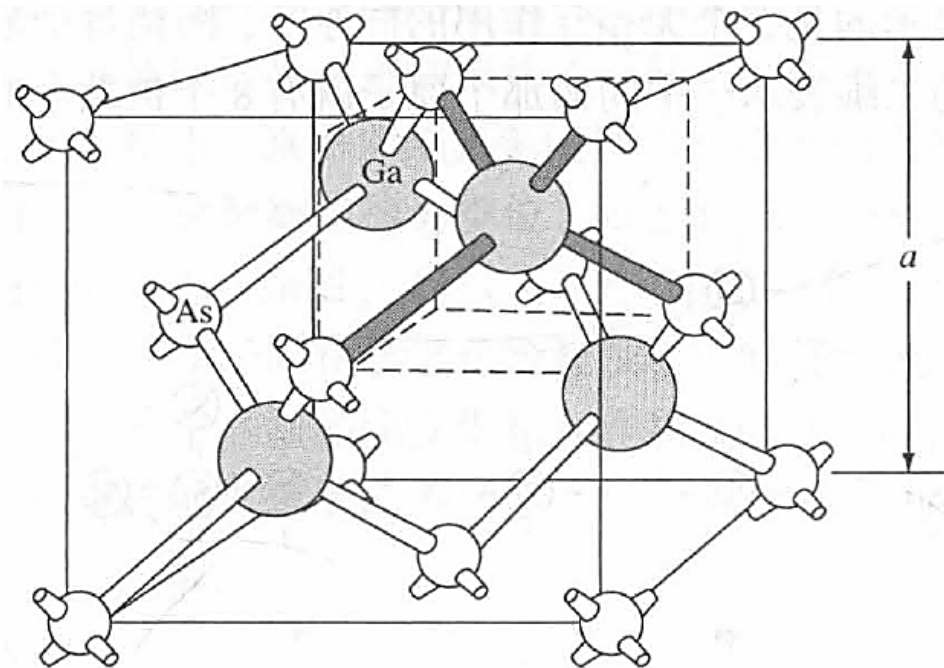


图 1.14 砷化镓的铅锌矿(闪锌矿)晶格

Home Work 1

4. 计算原子密度

(a) GaAs 的晶格常数为 $5.65\text{\AA}$

确定每立方厘米中 Ga 原子和 As 原子的数量。

(b) 确定锗半导体中锗原子的体密度。

锗的晶格常数为 $5.65\text{\AA}$ 。

4. (a) The lattice constant of GaAs is $5.65\text{\AA}$. Determine the number of Ga atoms and As atoms per cm^3 . (b) Determine the volume density of germanium atoms in a germanium semiconductor. The lattice constant of germanium is $5.65\text{\AA}$.

Answer:

(b) Ge has the diamond structure and has 8 Ge atoms per unit cell.

$$\text{Density} = \frac{8}{(5.65 \times 10^{-8})^3} = 4.44 \times 10^{22} [\text{atoms}/\text{cm}^3]$$

之前已经指出，硅是最常用的半导体材料。硅是IV族元素，具有金刚石晶格结构。锗也是IV族元素，它同样为金刚石结构。与目前已考虑过的简立方结构相比，图 1.11 所示的金刚石结构晶胞要复杂得多。

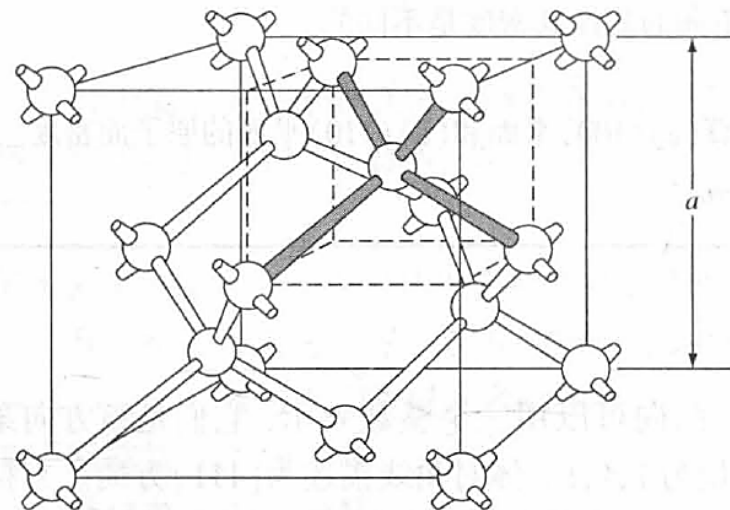


图 1.11 金刚石结构

Home Work 1

5. 原子占用体积百分比

假设每个原子是一个硬球，其表面与最近邻原子的表面接触。确定单位晶胞中所有原子体积占单位晶胞总体积的百分比：

- (a) 简单立方晶格
- (b) 面心立方晶格
- (c) 体心立方晶格
- (d) 金刚石晶格

(a) simple cubic

Assume that the lattice constant is a . Radius r of the sphere atom touching the nearest neighbor is $\frac{a}{2}$. Hence the volume of one atom V_{atom} is $\frac{4\pi}{3}(\frac{a}{2})^3$. Number of the atom n in the unit cell of cubic crystal is then 1 ($\frac{1}{8} \times 8$). The percentage of the total unit cell (density of atoms) $\%_{volume}$ can be obtained by

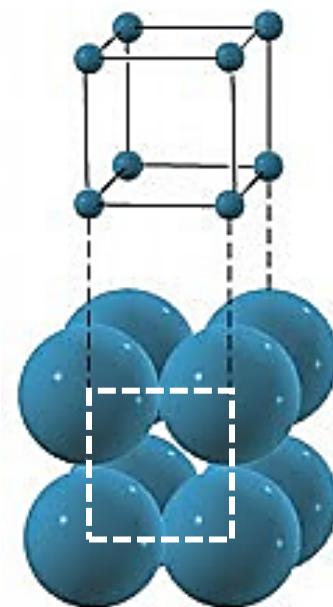
$$\%_{volume} = V_{atom} \times n / a^3 = \frac{\pi}{6} = 52.4\% \quad (1)$$

5. Assume that each atom is a hard sphere with the surface of each atom in contact with the surface of its nearest neighbor. Determine the percentage of volume of all atoms (volume of all atoms in a unit cell/volume of a unit cell) in a unit cell of hat is occupied in (a) a simple cubic lattice, (b) a face-centered cubic lattice, (c) a body-centered cubic lattice, and (d) diamond lattice.

Answer:

单胞内所有原子的体积/单胞的体积

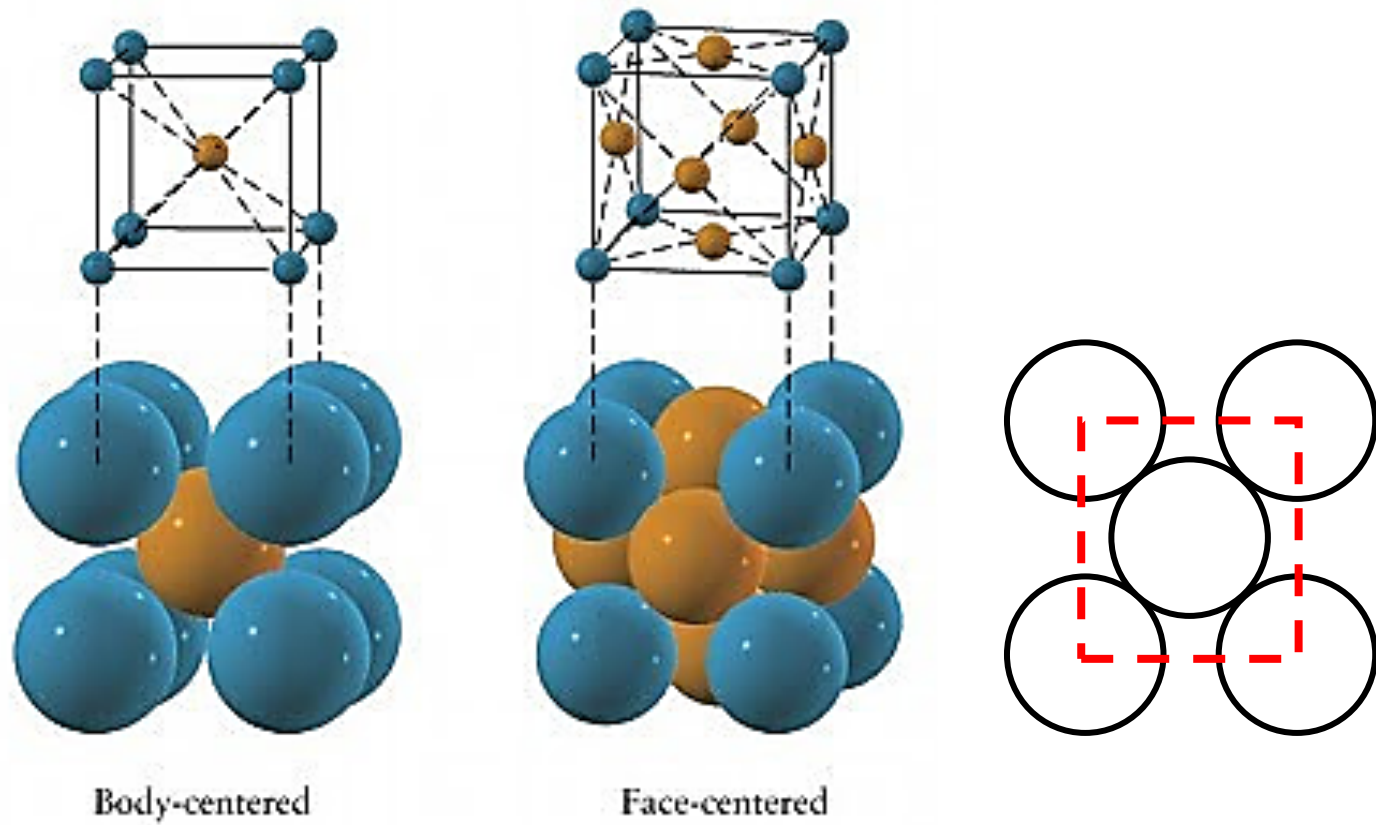
球体体积: $\frac{4}{3} \pi R^3$ a^3



Primitive
© 2007 Thomson Higher Education

Home Work 1

Answer:



Similarly, $r = \frac{\sqrt{2}}{4}a$, $n = 4$ (question 1.1 (a)). Therefore,

$$\%_{volume} = V_{atom} \times n/a^3 = \frac{4\pi}{3} \left(\frac{\sqrt{2}}{4}a \right)^3 \times 4/a^3 = 0.740 = 74\% \quad (2)$$

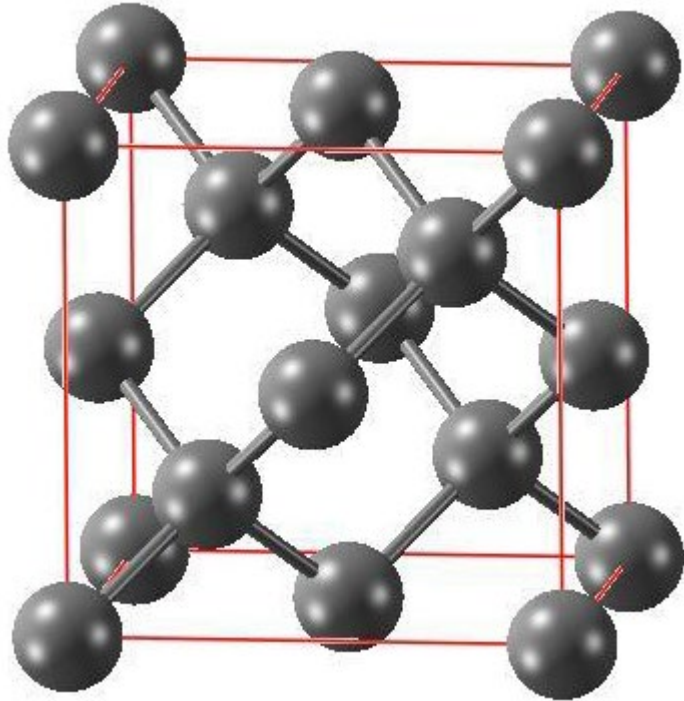
(c) bcc

$r = \frac{\sqrt{3}}{4}a$ and $n = 2$ (question 1.1 (b)). Therefore,

$$\%_{volume} = V_{atom} \times n/a^3 = \frac{4\pi}{3} \left(\frac{\sqrt{3}}{4}a \right)^3 \times 2/a^3 = 0.68 = 68\% \quad (3)$$

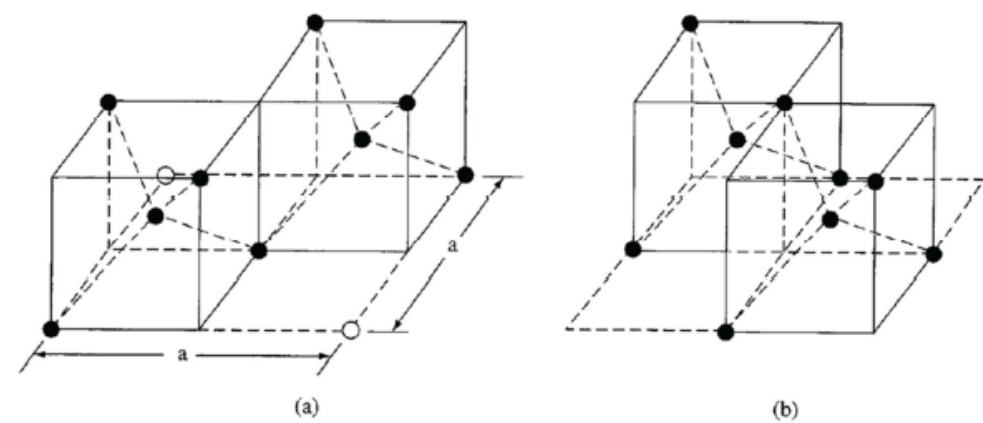
Home Work 1

Answer:

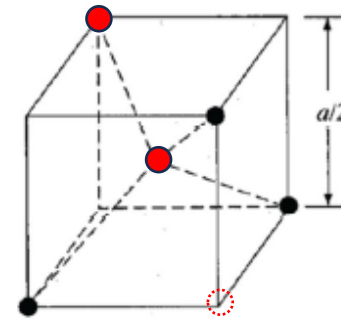


(d) diamond

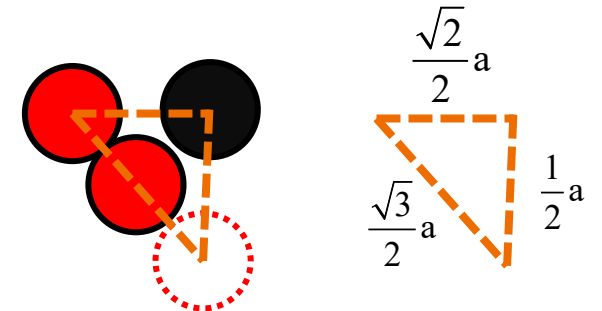
The smallest distance between the atoms is the distance between the corner and the middle in the "tetrahedron" (figure 1.11, page 10 of Neaman book). The half of the distance, that is the radius r , is $\frac{\sqrt{3}}{8}a$. Furthermore $n = 8$ (question 1.1 (b)).



Bottom half (a) and top half (b) portions of diamond lattice.



The tetrahedral structure of closest neighbors in the diamond lattice



$$\%_{volume} = V_{atom} \times n/a^3 = \frac{4\pi}{3} \left(\frac{\sqrt{3}}{8}a \right)^3 \times 8/a^3 = \frac{\pi\sqrt{3}}{16} = 34\% \quad (4)$$

Home Work 1

6. 硅的相关计算

如果硅的晶格常数为 $5.43\text{\AA}$ ，计算

(a) 从一个硅原子中心到其最近邻

原子中心的距离

(b) 硅原子的数量密度 (每 cm^3)

(c) 硅的质量密度 (克每 cm^3)

6. If the lattice constant of Si is $5.43\text{ \AA}$, calculate (a) the distance from the center of one Si atom to the center of its nearest neighbor, (b) the number density of Si atoms (per cm^3), and (c) the mass density (gram per cm^3) of Si.

$$a=5.43\text{\AA}=5.43 \times 10^{-8} \text{ cm}$$

(a) The smallest distance between atoms is $\frac{\sqrt{3}}{4}a$ (the answer for the question above). Using the lattice constant gives $a=2.35\text{\AA}$

(b) $(8 \text{ atoms}) / (\text{the volume of } (5.43 \times 10^{-8})^3) = 5.0 \times 10^{22} \text{ atoms/cm}^3$

(c) Mass per atom is $28.06 \text{ amu} = 4.66 \times 10^{-26} \text{ kg}$. Multiply the value with the density of atoms (the answer above) results 2.33 g/cm^3

Home Work 1

7. 三维立方晶格的平面与方向
考虑一个三维立方晶格，晶格常数为 a

(a) 绘制以下平面：(i) (100), (ii) (110), (iii) (310), (iv) (230)。

(b) 绘制以下方向：(i) [100], (ii) [110], (iii) [310], (iv) [230]。

7. Consider a three-dimensional cubic lattice with a lattice constant equal to a . (a) Sketch the following planes: (i) (100), (ii) (110), (iii) (310) and (iv) (230). (b) Sketch the following directions (i) [100], (ii) [110], (iii) [310] and (iv) [230].

Answer:

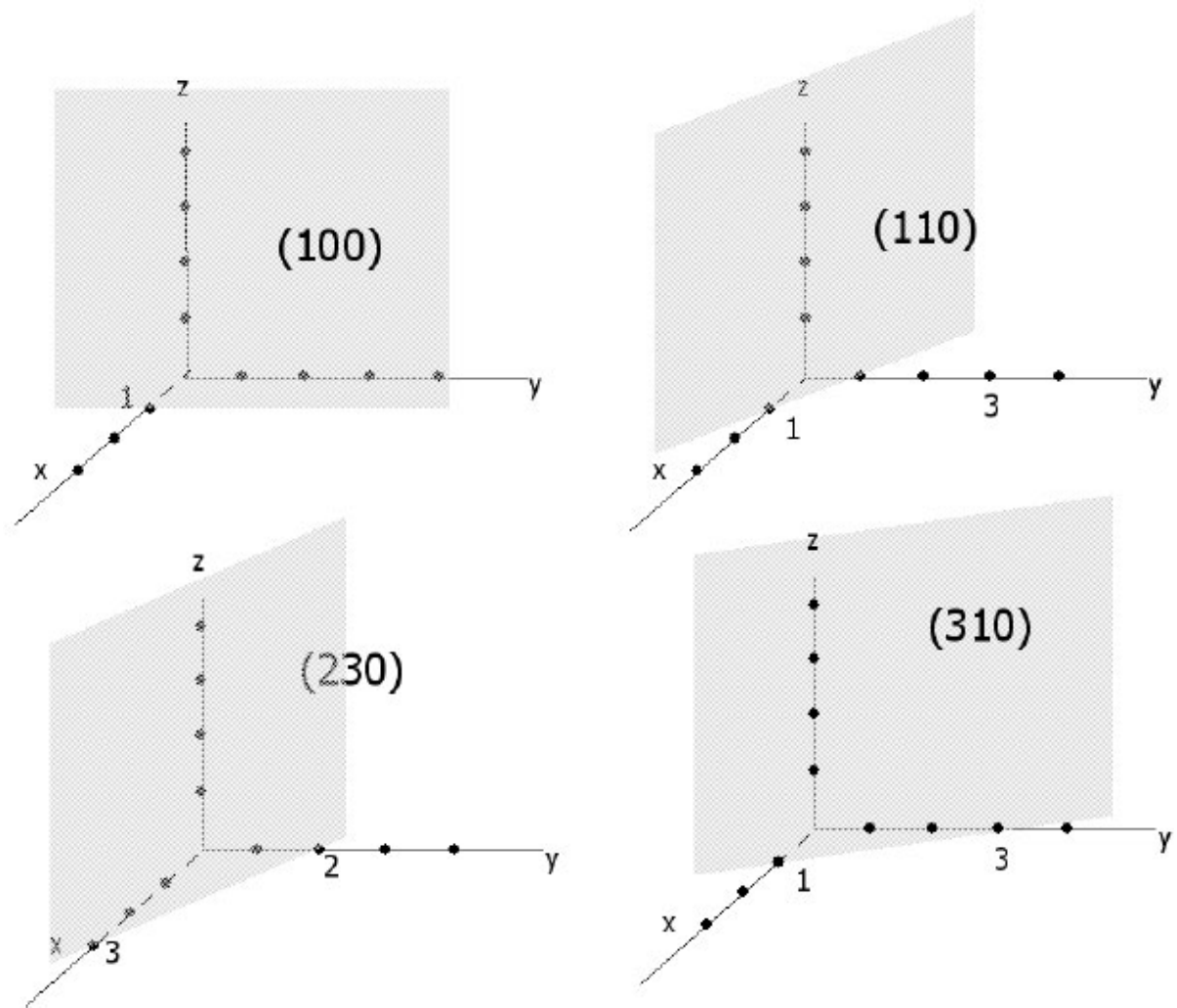


Figure 1: Crystal planes

Home Work 1

7. 三维立方晶格的平面与方向

考虑一个三维立方晶格，晶格

常数为 a

(a) 绘制以下平面：(i) (100), (ii)

(110), (iii) (310), (iv) (230)。

(b) 绘制以下方向：(i) [100], (ii)

[110], (iii) [310], (iv) [230]。

7. Consider a three-dimensional cubic lattice with a lattice constant equal to a . (a) Sketch the following planes: (i) (100), (ii) (110), (iii) (310) and (iv) (230). (b) Sketch the following directions (i) [100], (ii) [110], (iii) [310] and (iv) [230].

Answer:

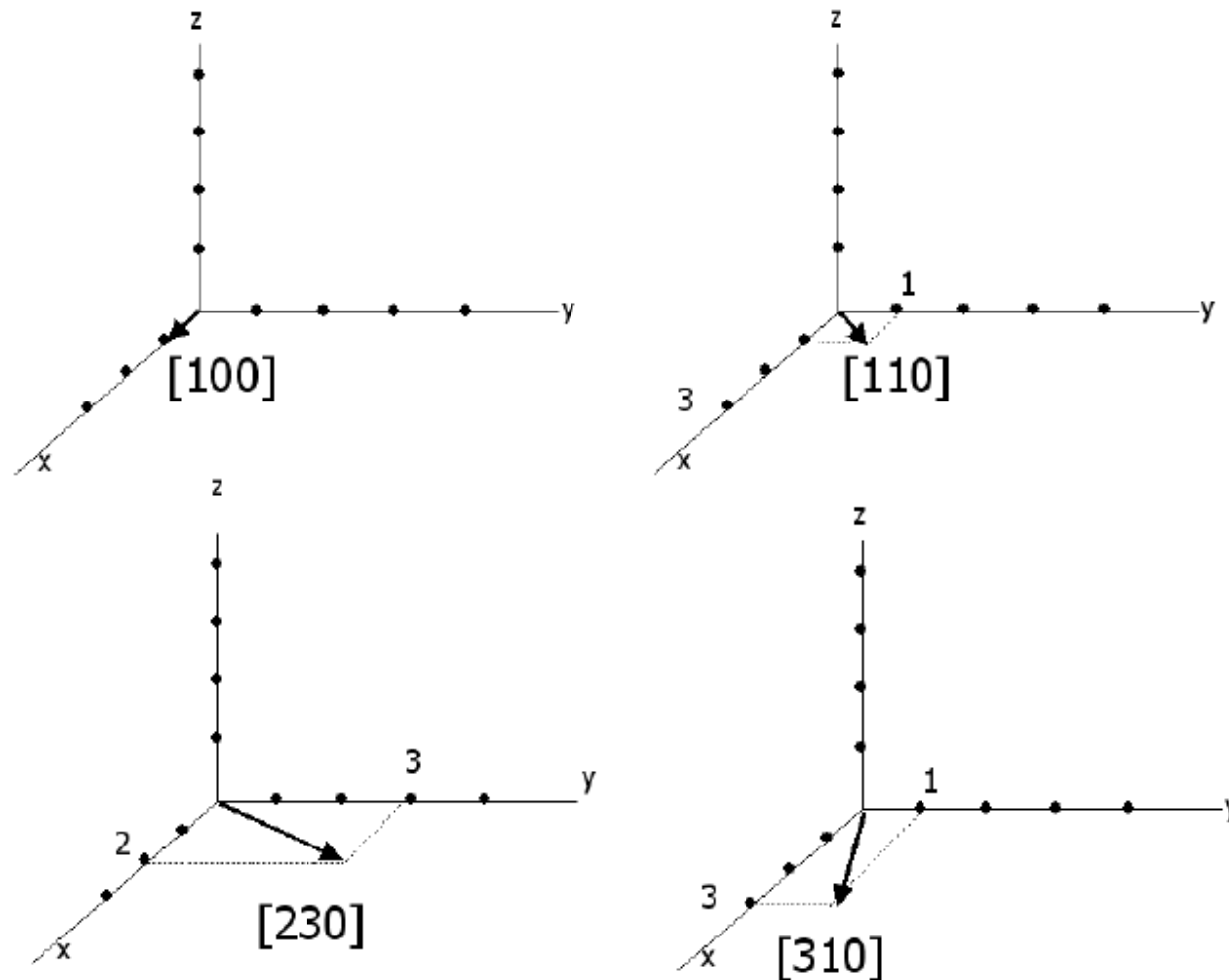


Figure 2: Crystal directions

Home Work 1

8. 硅的表面密度

确定硅在以下平面上的原子表面密度

(a) (100) 平面

(b) (110) 平面

(c) (111) 平面

8. Determine the surface density of atoms for Si on the (a) (100) plane, (b) (110) plane, and (c) (111) plane.

Answer:

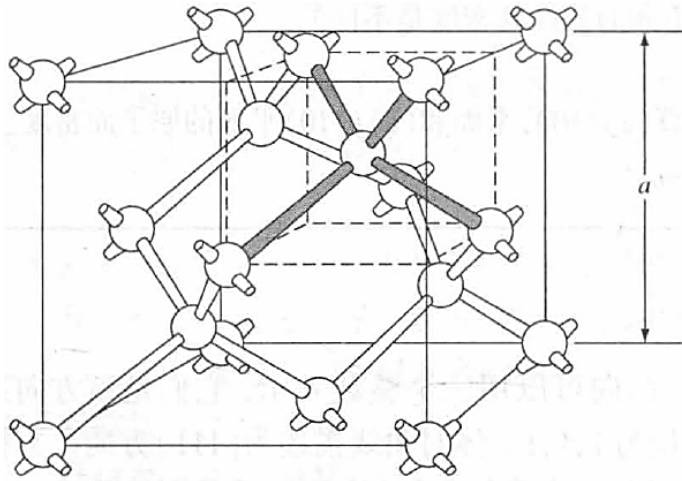
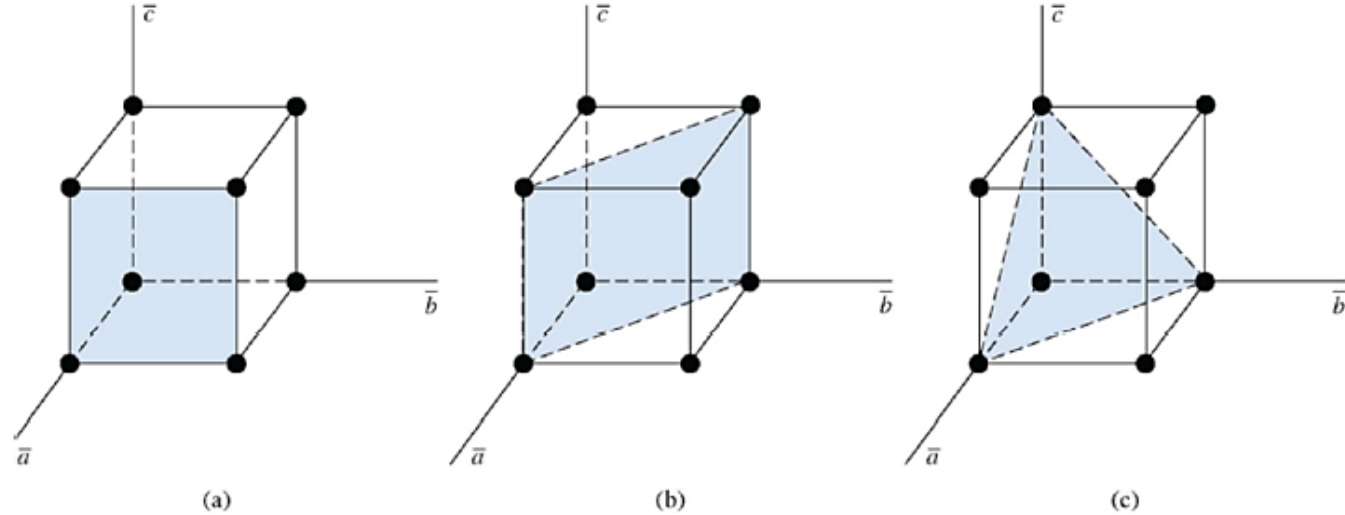


图 1.11 金刚石结构



Three lattice planes – (a) (100) plane, (b) (110) plane and (c) (111) plane

$$\text{面密度} = \frac{\text{每晶面的原子数}}{\text{每晶面的面积}}$$

题6给出，硅的晶格常数为 $5.43\text{\AA}$

Home Work 1

8. 硅的表面密度

确定硅在以下平面上的原子表面密度

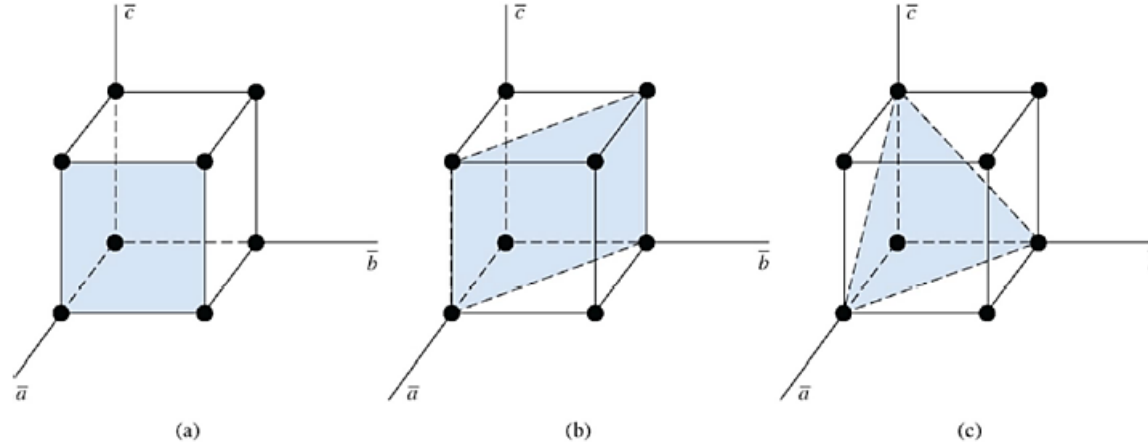
(a) (100) 平面

(b) (110) 平面

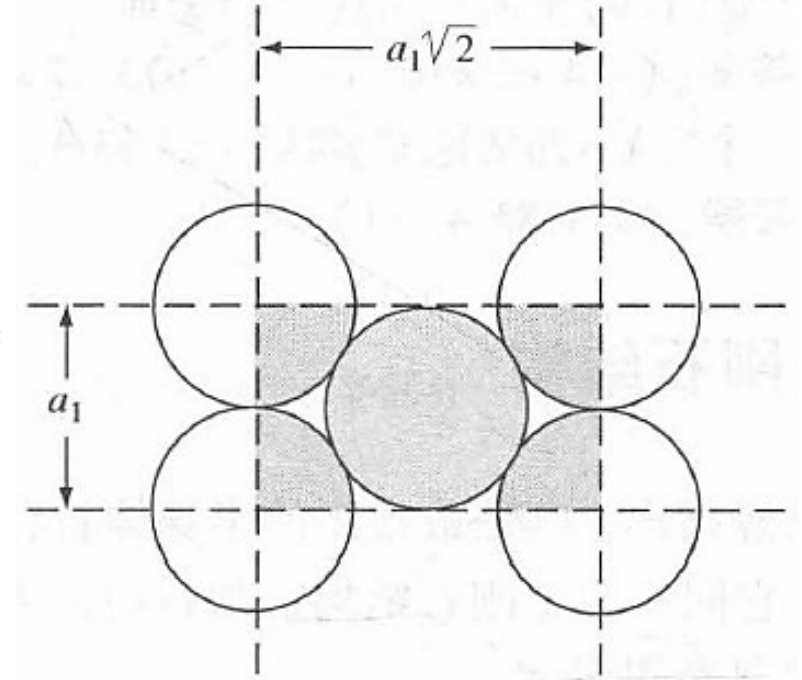
(c) (111) 平面

8. Determine the surface density of atoms for Si on the (a) (100) plane, (b) (110) plane, and (c) (111) plane.

Answer:



Three lattice planes – (a) (100) plane, (b) (110) plane and (c) (111) plane



(a) [100] plane of the Si has 2 atoms ($1/4 \times 4 + 1$) in the unit cell with the area of a^2 , where a is the lattice constant. Surface density D is then $2/(5.43 \times 10^{-8})^2 = 6.78 \times 10^{14} \text{ atoms/cm}^2$

Home Work 1

8. 硅的表面密度

确定硅在以下平面上的原子表面密度

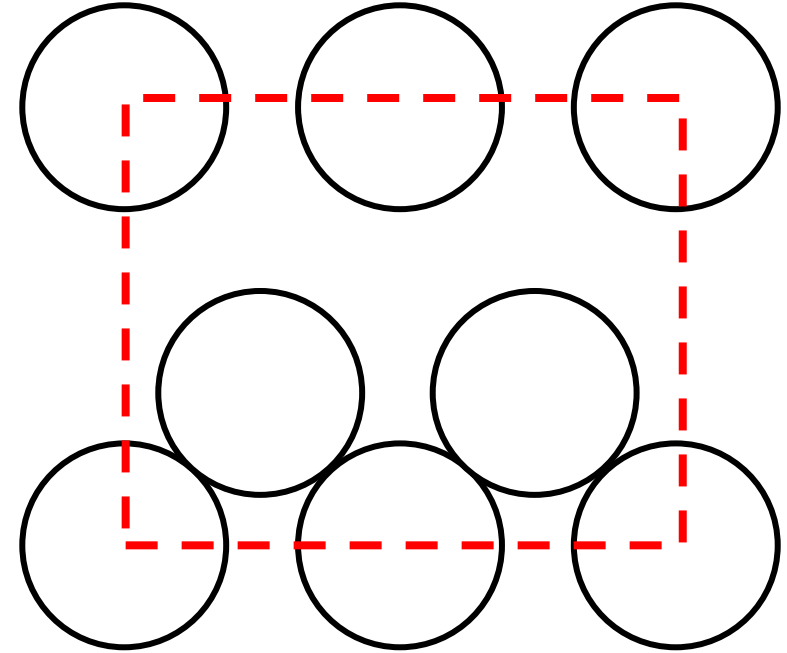
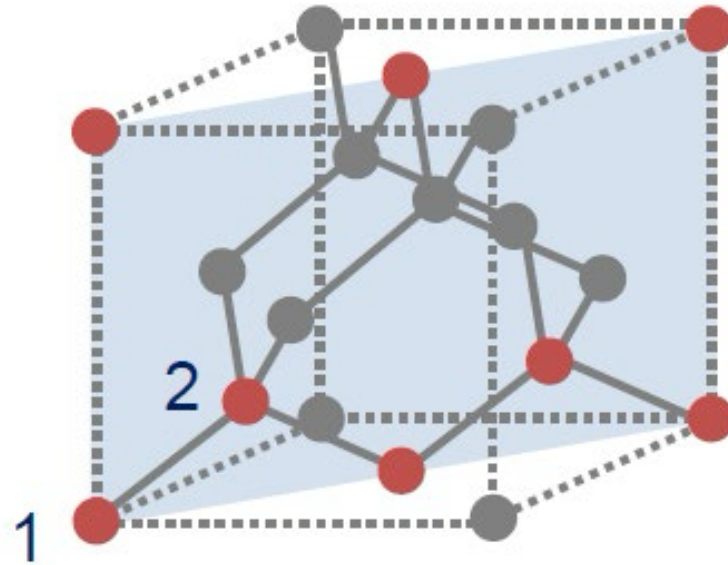
(a) (100) 平面

(b) (110) 平面

(c) (111) 平面

8. Determine the surface density of atoms for Si on the (a) (100) plane, (b) (110) plane, and (c) (111) plane.

Answer:



(b) [110] plane of the Si has 4 atoms $(2 + 1/2 \times 2 + 1/4 \times 4)$ in the unit cell with the area of $a \times \sqrt{2}a$.
Surface density D is then $4/(\sqrt{2} \times 357 \times 10^{-8})^2 = 9.59 \times 10^{14} \text{ atoms/cm}^2$

Home Work 1

8. 硅的表面密度

确定硅在以下平面
上的原子表面密度

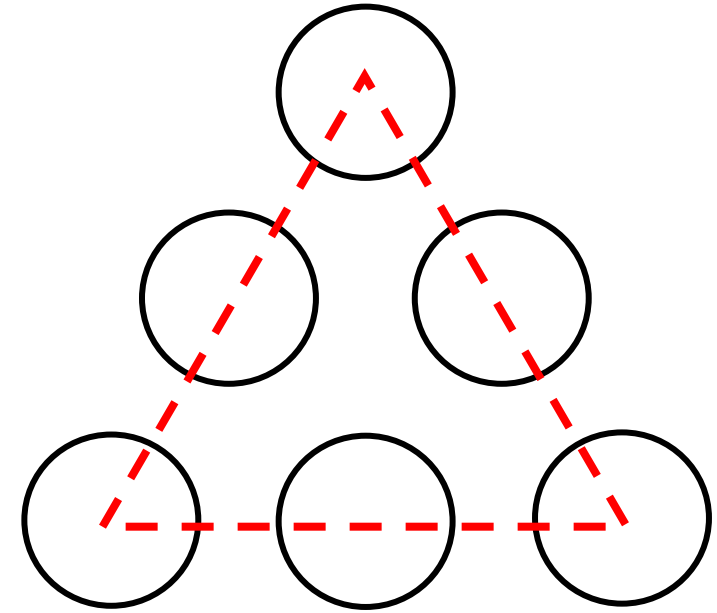
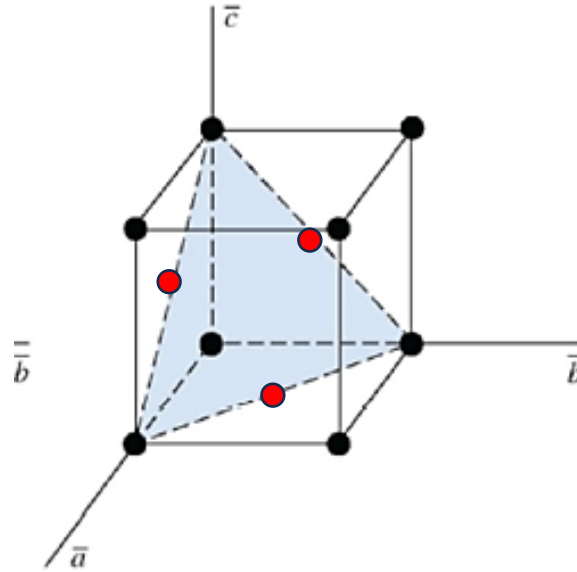
(a) (100) 平面

(b) (110) 平面

(c) (111) 平面

8. Determine the surface density of atoms for Si on the (a) (100) plane, (b) (110) plane, and (c) (111) plane.

Answer:



(c)[111] plane of the Si has 2 atoms ($3 \times 1/2 + 3 \times 1/6$) in the unit cell with the area of an equilateral (a regular) triangle with the side length of $\sqrt{2}a$. Surface density D is then $2/(1/2 \times \sqrt{3}a^2) = 7.83 \times 10^{14} \text{ atoms/cm}^2$

Home Work 2

1. 图1显示了两种特定半导体材料中导带内电子的抛物线E-k关系。确定这两个电子的有效质量（以自由电子质量 m 为单位）

1. Figure 1 shows the parabolic E versus k relationship in the conduction band for an electron in two particular semiconductor materials. Determine the effective mass (in units of the free electron mass m) of the two electrons.

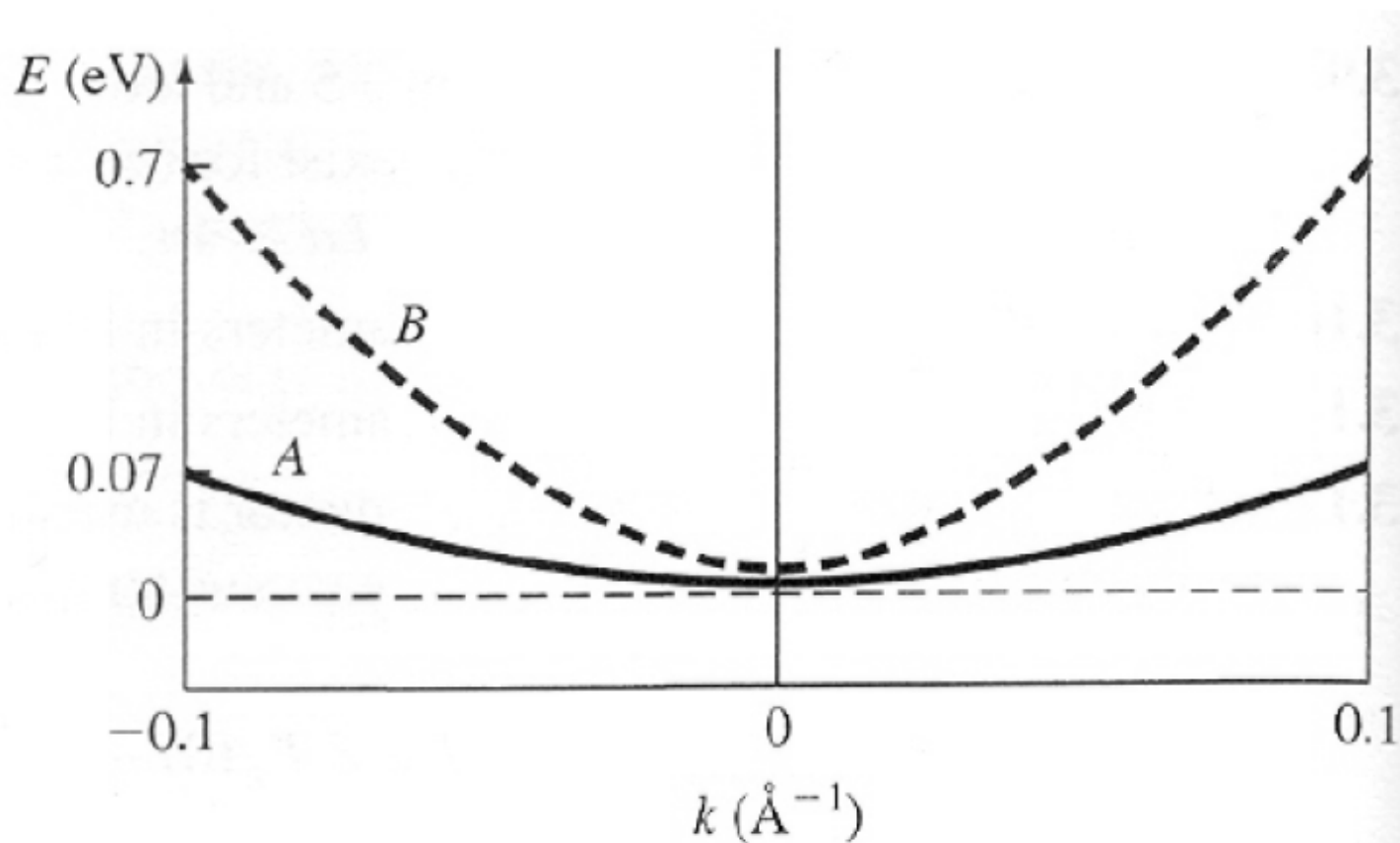


Figure 1: The E-k diagram

Home Work 2

Answer:

曲线 A 的计算

- 给定数据: $E = 0.07 \text{ eV}$, $k = 0.1 \text{ \AA}^{-1}$ 。
- 单位转换:
 - $k = 0.1 \text{ \AA}^{-1} = 0.1 \times 10^{10} \text{ m}^{-1} = 10^9 \text{ m}^{-1}$;
 - $E = 0.07 \text{ eV} = 0.07 \times 1.6 \times 10^{-19} \text{ J} = 1.12 \times 10^{-20} \text{ J}$ 。
- 代入公式:

$$1.12 \times 10^{-20} = \frac{(1.0545718 \times 10^{-34})^2 \times (10^9)^2}{2m^*}$$

计算 $\hbar^2 k^2$:

- $\hbar^2 = (1.0545718 \times 10^{-34})^2 \approx 1.112 \times 10^{-68} \text{ J}^2 \cdot \text{s}^2$;
- $k^2 = (10^9)^2 = 10^{18} \text{ m}^{-2}$;
- $\hbar^2 k^2 = 1.112 \times 10^{-68} \times 10^{18} = 1.112 \times 10^{-50} \text{ J}^2 \cdot \text{m}^2$;
- 代入:

$$1.12 \times 10^{-20} = \frac{1.112 \times 10^{-50}}{2m^*}$$

$$1.12 \times 10^{-20} = \frac{5.56 \times 10^{-51}}{m^*}$$

解出 m^* :

$$m^* = \frac{5.56 \times 10^{-51}}{1.12 \times 10^{-20}} = 4.96 \times 10^{-31} \text{ kg}$$

- 与自由电子质量比较:
 - 自由电子质量 $m_0 = 9.11 \times 10^{-31} \text{ kg}$;
 - 因此, $\frac{m_A}{m_0} = \frac{4.96 \times 10^{-31}}{9.11 \times 10^{-31}} = 0.544$ 。

$$E = \frac{p^2}{2m} = \frac{k^2 \hbar^2}{2m} \quad \text{能量与波矢关系式}$$

曲线 B 的计算

- 给定数据: $E = 0.7 \text{ eV}$, $k = 0.1 \text{ \AA}^{-1}$ 。
- 单位转换:
 - $k = 10^9 \text{ m}^{-1}$;
 - $E = 0.7 \text{ eV} = 0.7 \times 1.6 \times 10^{-19} \text{ J} = 1.12 \times 10^{-19} \text{ J}$ 。
- 代入公式:

$$1.12 \times 10^{-19} = \frac{1.112 \times 10^{-50}}{2m^*}$$

$$1.12 \times 10^{-19} = \frac{5.56 \times 10^{-51}}{m^*}$$

解出 m^* :

$$m^* = \frac{5.56 \times 10^{-51}}{1.12 \times 10^{-19}} = 4.96 \times 10^{-32} \text{ kg}$$

- 与自由电子质量比较:
 - $\frac{m_B}{m_0} = \frac{4.96 \times 10^{-32}}{9.11 \times 10^{-31}} = 0.0544$ 。

Home Work 2

Answer: (I.1)

Solution

E versus k has a parabolic relationship.

$$E_V - E = \frac{k^2 \hbar^2}{2m} \quad (1)$$

$$\underline{k = 0.1[\text{\AA}^{-1}] = 10^9[\text{m}^{-1}]} \quad (2)$$

For the curve A:

$$(0.07)(1.6 \times 10^{-19}) = \frac{(10^9)^2 (1.054 \times 10^{-34})^2}{2m} \quad (3)$$

which yields

$$m_A = 4.96 \times 10^{-31}[\text{kg}] \rightarrow \underline{\frac{m_A}{m_0} = 0.544} \quad (4)$$

For the curve B:

$$(0.7)(1.6 \times 10^{-19}) = \frac{(10^9)^2 (1.054 \times 10^{-34})^2}{2m} \quad (5)$$

which yields

$$m_B = 4.96 \times 10^{-32}[\text{kg}] \rightarrow \underline{\frac{m_B}{m_0} = 0.0544} \quad (6)$$

Home Work 2

2. 图2显示了一个特定允许能带的E-k关系图。确定(a)有效质量的符号和(b)图中四个位置处粒子的速度方向。

2. The E versus k diagram for a particular allowed energy band is shown in Figure 2. Determine (a) the sign of the effective mass and (b) the direction of velocity for a particle at each of the four positions shown.

$$p = \hbar k$$

$$\frac{\hbar^2 k^2}{2m}$$

$$\frac{1}{\hbar} \frac{dE}{dk} = \frac{p}{m} = \boxed{v}$$

$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{1}{m}$$

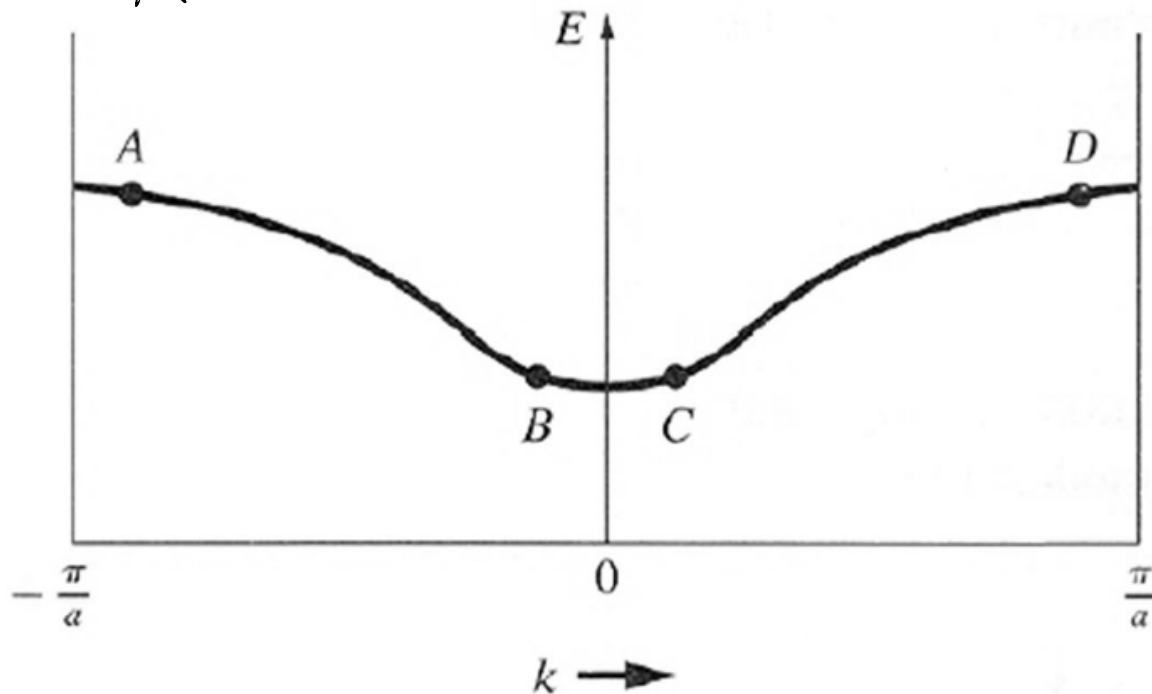


Figure 2: The E-k diagram in a band

Home Work 2

Answer:

- 速度方向:

粒子的速度 v 与能量对波矢的导数相关:

$$v = \frac{1}{\hbar} \frac{\partial E}{\partial k}$$

- 如果 $\frac{\partial E}{\partial k} > 0$, 则 $v > 0$ (正方向);
- 如果 $\frac{\partial E}{\partial k} < 0$, 则 $v < 0$ (负方向)。

- 有效质量符号:

有效质量 m^* 与能量对波矢的二阶导数相关:

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k^2}$$

- 如果 $\frac{\partial^2 E}{\partial k^2} > 0$ (曲线向上凸), 则 $m^* > 0$;
- 如果 $\frac{\partial^2 E}{\partial k^2} < 0$ (曲线向下凸), 则 $m^* < 0$ 。

分析每个点

- 点 A:

- $\frac{\partial E}{\partial k} < 0$ (斜率负), 因此 $v < 0$;
- $\frac{\partial^2 E}{\partial k^2} < 0$ (曲线向下凸), 因此 $m^* < 0$ 。

- 点 B:

- $\frac{\partial E}{\partial k} < 0$ (斜率负), 因此 $v < 0$;
- $\frac{\partial^2 E}{\partial k^2} > 0$ (曲线向上凸), 因此 $m^* > 0$ 。

- 点 C:

- $\frac{\partial E}{\partial k} > 0$ (斜率正), 因此 $v > 0$;
- $\frac{\partial^2 E}{\partial k^2} > 0$ (曲线向上凸), 因此 $m^* > 0$ 。

- 点 D:

- $\frac{\partial E}{\partial k} > 0$ (斜率正), 因此 $v > 0$;
- $\frac{\partial^2 E}{\partial k^2} < 0$ (曲线向下凸), 因此 $m^* < 0$ 。

Home Work 2

Answer:

Points A, B: $\frac{\partial E}{\partial k} < 0$, which means velocity in -x direction

Points C, D: $\frac{\partial E}{\partial k} > 0$, which means velocity in +x direction

Points A, D: $\frac{\partial^2 E}{\partial k^2} < 0$, which means negative effective mass;

Points B, C: $\frac{\partial^2 E}{\partial k^2} > 0$, which means positive effective mass.

Home Work 2

3. 考虑一个宽度为 a 的二维空间中的自由电子。允许的波数 k_x 和 k_y 分别表示为 $\pi n_x/a$ 和 $\pi n_y/a$ ，其中 $n_x, n_y = 1, 2, 3, \dots$ 图3显示了 k 空间中允许量子态的二维阵列示意图。

- (1) 确定一个允许 k 点在 k 空间中占据的面积。
- (2) 表示半径为 k 和 $k+dk$ 的第一象限圆所包围的区域内允许态的数量，其中 $k \gg dk$ 。
- (3) 表示实空间中每单位能量的允许态密度。

3. Consider free electron in a 2D box with a width of a . The allowed wave-numbers k_x and k_y for x and y directions respectively are expressed as $\pi n_x/a$ and $\pi n_y/a$ where $n_x, n_y = 1, 2, 3, \dots$. The figure 3 shows the schematic view of the 2D array of allowed quantum states in k -space.

- (1) Determine the area in k -space occupied by one allowed k point.
- (2) Express number of allowed states in the area enclosed by the first quadrant circles with the radius of k and $k + dk$, where $k \gg dk$.
- (3) Express density of allowed states in real space per unit energy.

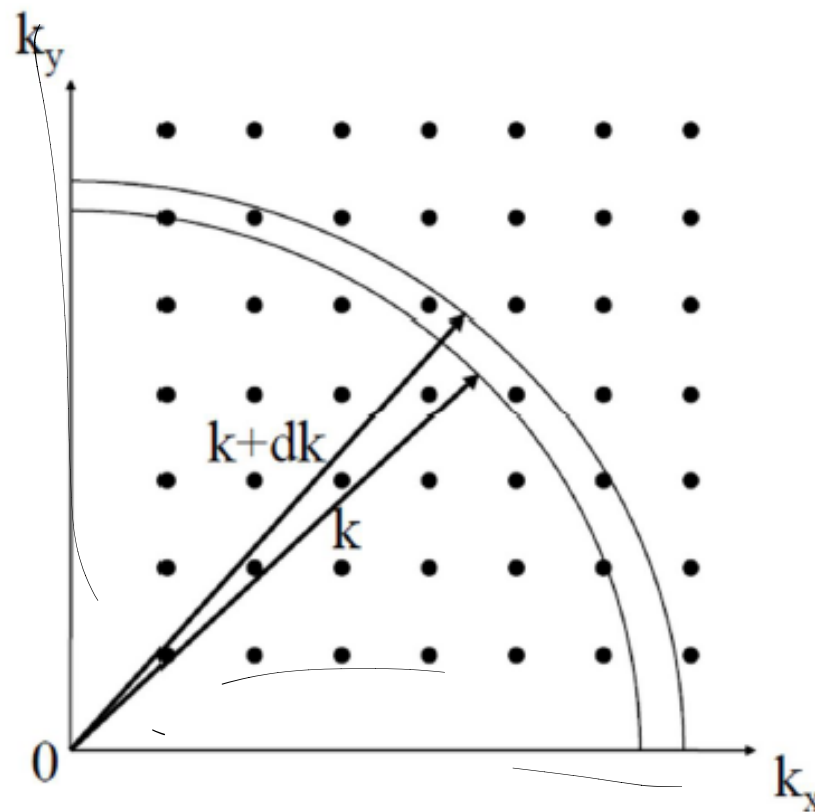


Figure 3. 2D array of allowed quantum states in k -space

Home Work 2

Answer:

(1) 每个允许 k 点的 k -空间面积

- k_x 和 k_y 的间隔分别为 $\Delta k_x = \frac{\pi}{a}$, $\Delta k_y = \frac{\pi}{a}$ 。
- 因此, 每个量子态在 k -空间中的面积为:

$$\Delta k_x \cdot \Delta k_y = \left(\frac{\pi}{a}\right) \cdot \left(\frac{\pi}{a}\right) = \left(\frac{\pi}{a}\right)^2$$

(2) 在第一象限中, 半径为 k 和 $k+dk$ 之间的区域内的允许态数

- 第一象限的圆环面积为:

$$\text{面积} = \frac{1}{4} \cdot 2\pi k \cdot dk = \frac{\pi k dk}{2}$$

- 每个态的面积为 $\left(\frac{\pi}{a}\right)^2$, 因此态数为:

$$\text{态数} = \frac{\frac{\pi k dk}{2}}{\left(\frac{\pi}{a}\right)^2} = \frac{\pi k dk}{2} \cdot \frac{a^2}{\pi^2} = \frac{a^2 k dk}{2\pi}$$

- 考虑自旋 (每个态有两个自旋状态), 总态数为:

$$2 \cdot \frac{a^2 k dk}{2\pi} = \frac{a^2 k dk}{\pi}$$

- 能量与波矢关系为 $E = \frac{\hbar^2 k^2}{2m}$, 求导得:

$$dE = \frac{\hbar^2 k}{m} \cdot dk \Rightarrow dk = \frac{m \cdot dE}{\hbar^2 k}$$

- 从步骤 2, 态数为 $\frac{a^2 k \cdot dk}{\pi}$, 代入 dk :

$$\text{态数} = \frac{a^2 k \cdot \frac{m \cdot dE}{\hbar^2 k}}{\pi} = \frac{a^2 m \cdot dE}{\pi \hbar^2}$$

- 总态数为 $g(E) \cdot dE \cdot A$, 其中 $A = a^2$ 为盒子面积:

$$g(E) \cdot dE \cdot a^2 = \frac{a^2 m \cdot dE}{\pi \hbar^2}$$

- 解得:

$$g(E) = \frac{m}{\pi \hbar^2}$$

- 实空间单位面积下的态密度 $g_{2D}(E) = \frac{g(E)}{A} = \frac{m}{\pi \hbar^2}$ 。

Home Work 2

Answer:

(1) Area occupied by 1 k value is

$$\underline{\left(\frac{\pi}{a}\right)^2} \quad (1)$$

(2) Number of allowed states per unit area in k -space is then

$$2 \times \left(\frac{a}{\pi}\right)^2, \quad (2)$$

where the spin effect is taken into account.

Area enclosed by the first quadrant circles with the radius k and $k + dk$ can be expressed by

$$\frac{1}{4}\pi(k + dk)^2 - \frac{1}{4}\pi k^2 = \frac{1}{4} \cdot 2\pi k \cdot dk \quad (3)$$

Number of allowed states in the area enclosed by the first quadrant circles with the radius k and $k + dk$ can be expressed by

$$\underline{2 \times \left(\frac{a}{\pi}\right)^2 \cdot \frac{1}{4} \cdot 2\pi k \cdot dk = \frac{a^2}{\pi} k \cdot dk} \quad (4)$$

Home Work 2

Answer:

(3) The relation between k and E is

$$k = \frac{1}{\hbar} \sqrt{2mE}. \quad (5)$$

Taking the differential, we obtain

$$dk = \frac{1}{\hbar} \sqrt{2m} \frac{1}{2\sqrt{E}} dE = \frac{1}{\hbar} \sqrt{\frac{m}{2E}} dE. \quad (6)$$

Substituting these expressions into the equation 4, we obtain

$$\frac{a^2}{\pi} \frac{1}{\hbar} \sqrt{2mE} \cdot \frac{1}{\hbar} \sqrt{\frac{m}{2E}} dE = \frac{a^2 m}{\pi \hbar^2} dE, \quad (7)$$

which represents number of allowed states in real space per unit energy, between E and $E + \Delta E$. Density of allowed states in real space per unit energy can be obtained by deviding the equation by a^2 and dE , i.e.,

$$\frac{m}{\pi \hbar^2}, \quad (8)$$

Home Work 2

4. 确定硅 (Si) 中(a) E_c 到 E_c+kT 之间 ($T=300\text{ K}$) 和 (b) E_v 到 E_v-kT 之间 ($T=300\text{ K}$) 的总能量态数。

4. Determine the total number of energy states in Si (a) between E_c and $E_c + kT$ at $T = 300\text{K}$, and (b) between E_v and $E_v - kT$ at $T = 300\text{K}$.

- 导带的态密度:

$$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \sqrt{E - E_C}$$

- 总态数:

$$N_c = \int_{E_C}^{E_C+kT} g_c(E) dE = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \int_0^{kT} \sqrt{E} dE = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3} (kT)^{3/2}$$

- 类似地, 价带的总态数:

$$N_v = \frac{4\pi(2m_p^*)^{3/2}}{h^3} \cdot \frac{2}{3} (kT)^{3/2}$$

- 使用硅的参数:

- $m_n^* = 1.08m_0$;
- $m_p^* = 0.56m_0$;
- $kT = 0.0259\text{ eV}$ at 300 K .

- 计算结果:

- $N_c = 2.12 \times 10^{19} \text{ cm}^{-3}$;
- $N_v = 7.92 \times 10^{18} \text{ cm}^{-3}$.

Home Work 2

Answer:

(1.2) (a)

$$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \sqrt{E - E_C}$$

Then

$$g_T = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \int_{E_C}^{E_C + kT} (E - E_C)^{1/2} dE = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \left(\frac{2}{3} \right) (E - E_C)^{3/2} \Big|_{E_C}^{E_C + kT}$$

or

$$g_T = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \left(\frac{2}{3} \right) (kT)^{3/2}$$

which yields

$$g_T = 2.12 \times 10^{25} \text{m}^{-3} = 2.12 \times 10^{19} \text{cm}^{-3}$$

(b) Similarly,

$$g_T = \frac{4\pi(2m_p^*)^{3/2}}{h^3} \left(\frac{2}{3} \right) (kT)^{3/2}$$

$$g_T = 7.92 \times 10^{24} \text{m}^{-3} = 7.92 \times 10^{18} \text{cm}^{-3}$$

Home Work 2

5. 假设费米能级恰好位于半导体带隙能量的中心，温度为 $T=300\text{ K}$ 。(a) 计算硅 (Si) 和锗 (Ge) 中导带底能量态被电子占据的概率。(Si的 $E_g=1.12\text{ eV}$ ，Ge的 $E_g=0.66\text{ eV}$) (b) 计算硅 (Si) 和锗 (Ge) 中价带顶能量态为空穴的概率。

5. Assume the Fermi energy level is exactly in the center of the bandgap energy of a semiconductor at $T = 300\text{K}$.

(a) Calculate the probability that an energy state in the bottom of the conduction band is occupied by an electron for Si and Ge. ($E_g=1.12\text{ eV}$ for Si and $E_g=0.66\text{ eV}$ for Ge)

(b) Calculate the probability that an energy state in the top of the valence band is empty for Si and Ge.

• 费米-狄拉克分布：

$$f(E) = \frac{1}{1 + \exp\left(\frac{E-E_F}{kT}\right)}$$

• 对于导带底, $E = E_C$, 而 $E_F = \frac{E_C+E_V}{2}$, 因此 $E - E_F = \frac{E_g}{2}$ 。

• Si: $E_g = 1.12\text{ eV}$, $\frac{E_g}{2} = 0.56\text{ eV}$ 。

$$f(E_C) = \frac{1}{1 + \exp\left(\frac{0.56}{0.0259}\right)} = \frac{1}{1 + \exp(21.62)} \approx 4.07 \times 10^{-10}.$$

• Ge: $E_g = 0.66\text{ eV}$, $\frac{E_g}{2} = 0.33\text{ eV}$ 。

$$f(E_C) = \frac{1}{1 + \exp\left(\frac{0.33}{0.0259}\right)} = \frac{1}{1 + \exp(12.74)} \approx 2.93 \times 10^{-6}.$$

• 价带顶的空穴概率是 $1 - f(E_V)$, 但由于对称性, $f(E_V) = 1 - f(E_C)$, 因此空穴概率与导带底的占有概率相同。

Home Work 2

Answer:

(II.1) (a)

At $E = E_{\text{midgap}}$,

$$f(E) = \frac{1}{1 + \exp \frac{E - E_F}{kT}} = \frac{1}{1 + \exp \frac{E_g/2}{kT}}$$

Si: $E_g = 1.12 \text{ eV}$,

$$f(E) = \frac{1}{1 + \exp \frac{1.12}{2(0.0259)}} = \underline{4.07 \times 10^{-10}}$$

Ge: $E_g = 0.66 \text{ eV}$,

$$f(E) = \frac{1}{1 + \exp \frac{0.66}{2(0.0259)}} = \underline{2.93 \times 10^{-6}}$$

(b) The answers are the exactly the same as (a).

Home Work 2

6. Discuss the differences of metal, semiconductor and insulator in terms of (a) energy band (b) electrical and (c) optical properties.

6. 讨论金属、半导体和绝缘体在以下方面的区别：(a) 能带，(b) 电学性质，(c) 光学性质。

(a) 能带特性

- **金属**：在金属中，导带和价带重叠或没有能隙。这意味着电子可以自由移动，Fermi能级位于导带中，导致高电导率。
- **半导体**：半导体具有一个较小的禁带（通常0.1到3 eV），如硅（Si）的禁带约为1.1 eV。在室温下，少量电子可通过热激发从价带跃迁到导带，产生载流子（电子和空穴）。
- **绝缘体**：绝缘体的禁带宽度较大（>3 eV），如二氧化硅（SiO₂）的禁带约9 eV。在室温下，几乎没有电子能跨越禁带，因此没有自由载流子，电导率极低。

(b) 电学特性

- **金属**：由于导带中有大量自由电子，金属具有高电导率，例如铜的电阻率仅为 $1.68 \times 10^{-8} \Omega \cdot m$ 。适合用于导线和电路连接。
- **半导体**：半导体的电导率介于金属和绝缘体之间，可通过掺杂（如添加硼或磷）增加载流子浓度，从而显著提高导电性。纯硅的电导率较低，但掺杂后可用于制造晶体管。
- **绝缘体**：绝缘体的电导率极低，例如玻璃的电阻率可达 $10^{12} \Omega \cdot m$ 以上，常用作电线的绝缘层或电路隔离。

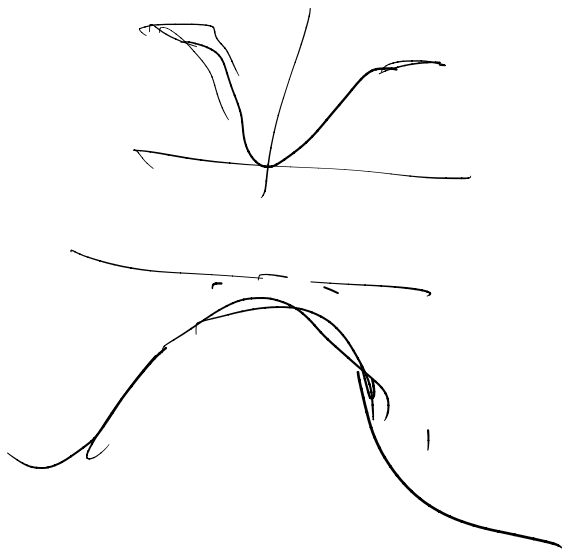
(c) 光学特性

- **金属**：金属对光具有高反射率，这是由于自由电子密度高，可以吸收并反射电磁波。例如，银镜反应中银的反射率接近100%。
- **半导体**：半导体可以吸收能量大于禁带宽度的光子，产生电子-空穴对。例如，GaAs的禁带为1.43 eV，可吸收红光以上波长的光，用于太阳能电池。
- **绝缘体**：如果禁带宽度大于可见光光子的能量（约1.7到3.1 eV），绝缘体对可见光透明，如玻璃；若禁带较小，可能吸收紫外光而显得不透明。

开放题无答案
合理即可

Home Work 2

7. 在什么情况下电子的有效质量 m 满足(a) $m/m_0 > 1$, (b) $m^*/m_0 < 1$, (c) $m^*/m_0 < 0$ 。



开放题无答案
合理即可

7. Under what situation that the effective mass of electron m^* (a) $m^*/m_0 > 1$, (b) $m^*/m_0 < 1$, (c) $m^*/m_0 < 0$.

有效质量来源于能带结构的曲率，具体来说，通过能量的二阶导数与波矢 k 的关系定义：

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E}{dk^2}$$

其中 $\hbar$ 是约化普朗克常数。能带的形状决定了 m^* 是大于、小于还是负于电子的静止质量 m_0 ($m_0 = 9.11 \times 10^{-31} \text{ kg}$)。

- 当能带曲率较小（即“平坦”）， m^* 较大。
- 当能带曲率较大（即“陡峭”）， m^* 较小。
- 当能带向下弯曲（如价带顶部）， m^* 为负。

情况 (a) : $m^*/m_0 > 1$

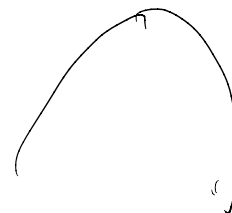
这种情况发生在电子的有效质量大于其静止质量，通常见于某些半导体的空穴。例如，在硅 (Si) 300K 时，空穴的有效质量为 $1.15 m_0$ ，大于1。此外，在一些特殊材料如重费米子系统中，有效质量可能达到 m_0 的1000倍，可能是由于电子与晶格的强相互作用。

情况 (b) : $m^*/m_0 < 1$

这种情况常见于半导体的电子，有效质量小于静止质量，表明电子在材料中的移动性更高。例如，砷化镓 (GaAs) 的电子有效质量为 $0.067 m_0$ ，锑化铟 (InSb) 为 $0.013 m_0$ ，都远小于1，这与导带曲率较陡有关。

情况 (c) : $m^*/m_0 < 0$

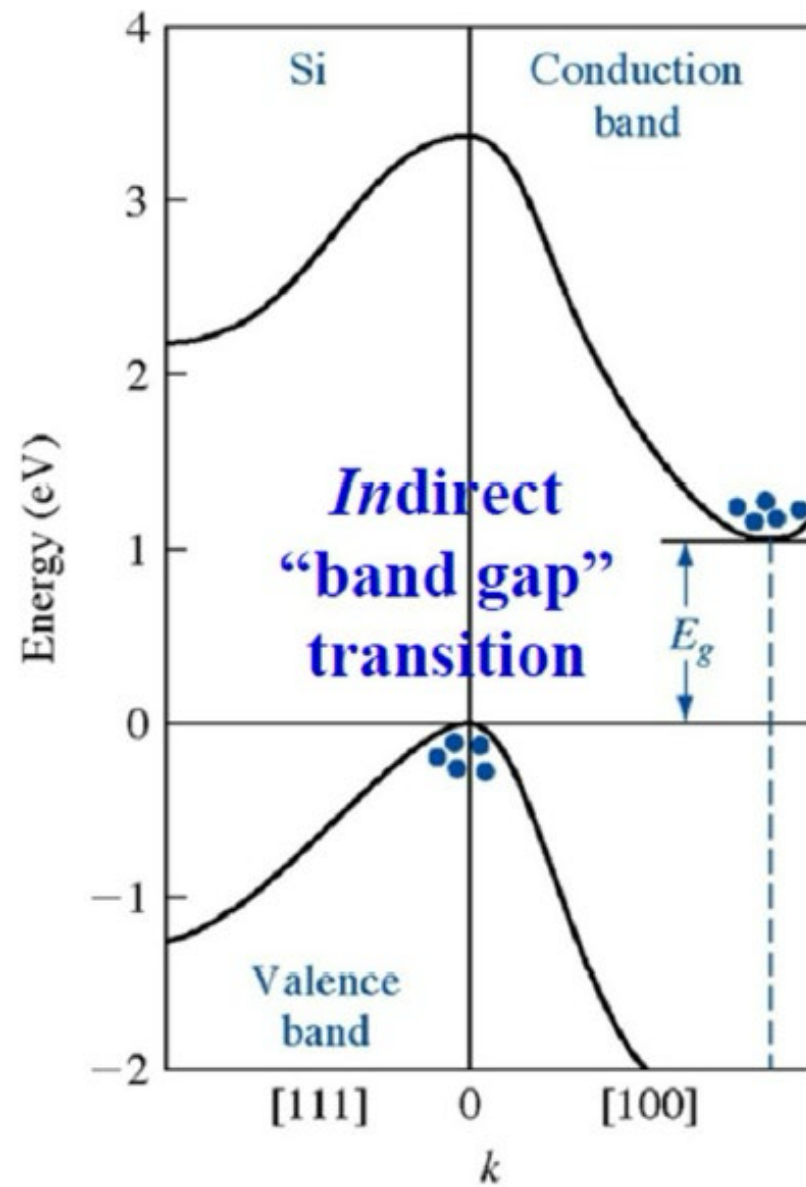
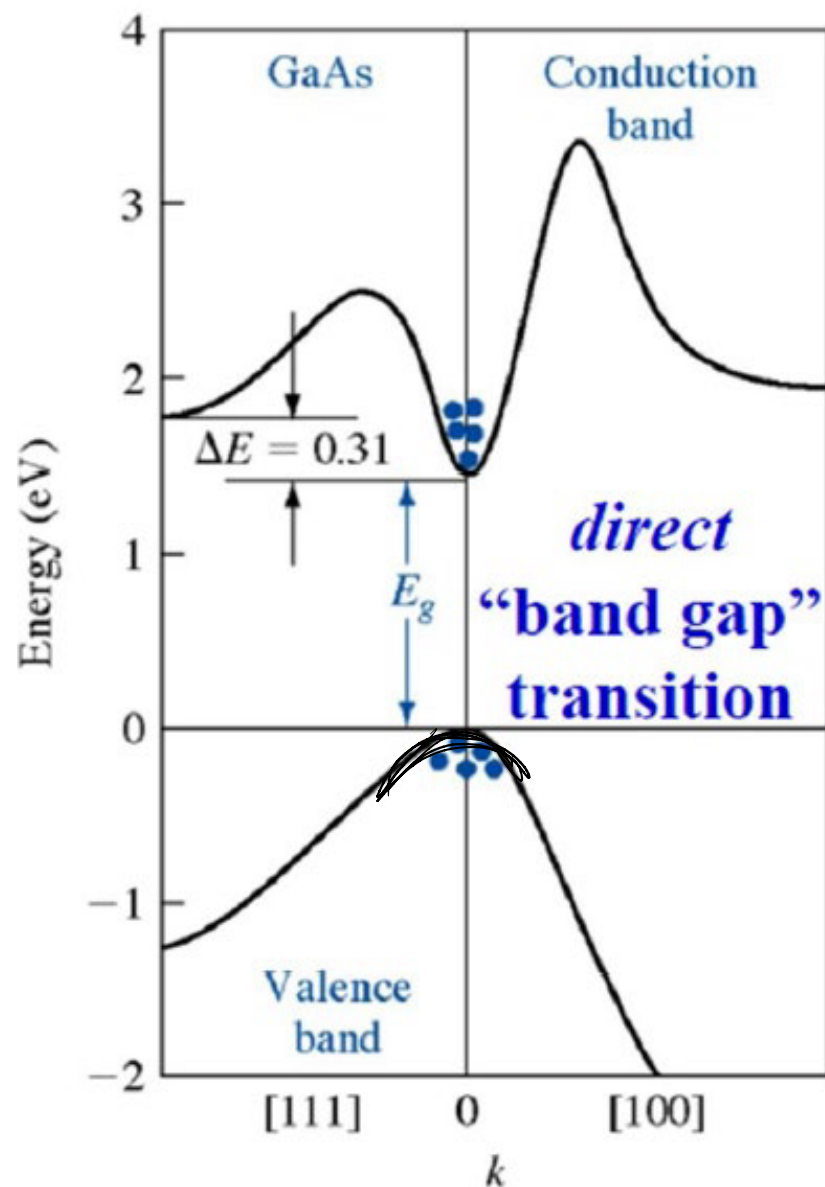
当有效质量为负时，发生在价带顶部的电子附近，此时能带向下弯曲。负有效质量意味着电子对力的响应如同带正电荷，这解释了半导体中的空穴概念。例如，在硅或锗的价带顶部，电子可能表现出负有效质量。



Home Work 2

8. 讨论直接带隙半导体和间接带隙半导体在以下方面的区别：(a) E-k图，(b) 应用

8. Discuss the differences of direct band-gap semiconductors and indirect band-gap semiconductors in terms of (a) E-K diagrams and (b) applications.



开放题无答案
合理即可

Home Work 2

Answer:

(a) E-K 图（能量-波矢图）

- **直接带隙半导体**：导带的最低点和价带的最高点位于 Brillouin 区的同一 k 点。例如，砷化镓（GaAs）的导带底和价带顶均在 Γ 点（ $k=0$ ），光子跃迁无需改变动量，效率高。
- **间接带隙半导体**：导带的最低点和价带的最高点位于不同 k 点。例如，硅（Si）的导带底在 X 点，价带顶在 Γ 点，光子跃迁需伴随 phonon 吸收或发射，效率较低。

(b) 应用

- **直接带隙半导体**：由于光子跃迁效率高，适合用于光电子器件，如发光二极管（LED）、激光二极管和太阳能电池。例如，GaAs 用于红外激光器。
- **间接带隙半导体**：由于光发射效率低，不适合光电子器件，但因材料丰裕（如硅）和工艺成熟，广泛用于电子器件，如晶体管和集成电路。